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Seminar report On

A Software-based Mitigation against Inaudible Voice Attacks

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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By

Shreya Veeraraghav (U2103198)

Under the guidance of

Dr. Anita John

Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039
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CERTIFICATE

This is to certify that the seminar report entitled ***"A Software-based Mitigation against Inaudible Voice Attacks"*** *is a bonafide record of the work done by* ***Shreya Veeraraghav (U2103198)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2024-2025.*

Dr. Anita John
Seminar Guide
Associate Professor
Dept. of CSE
RSET

Ms. Liya Joseph
Seminar Coordinator
Assistant Professor
Dept. of CSE
RSET

Dr. Preetha K G
Professor & HoD
Dept. of CSE
RSET

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Abstract

Speech recognition (SR) systems such as Siri, Google Now, Amazon Echo or Alexa have become an increasingly popular human-computer interaction method, and have turned various systems into voice controllable systems (VCS). Prior work shows that such systems are prone to attacks involving hidden commands, which are incomprehensible to people but not to these systems. This results in the manipulation of voice-controlled devices without the user knowing. The paper, "Learning Normality is Enough: A Software-based Mitigation against Inaudible Voice Attacks", proposes NormDetect, a universal software-based mitigation that can be instantly applied to devices without requiring any hardware modification as a mitigation method for such threats. The paper discovered that the patterns of inaudible voice attacks are diverse whereas benign audios share similar patterns hence the model uses an unsupervised-learning algorithm trained only on benign audios("The Normality") to detect attacks (the anomaly). The NormDetect maps audio spectrum features to a low-dimensional space called the latent space, performs similarity queries, and then reconstructs the spectrum from the latent space. This results in a significant reconstruction error for attacks than normality. The model's performance shows an average AUC of 99.48% and EER of 2.23%, suggesting the effectiveness of NormDetect in detecting inaudible voice attacks unlike existing studies' methods, in addition to being hardware-independent and attack-sample independent.

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List of Abbreviations

VCS - Voice Controlled System

MEMS - Micro Electro-Mechanical Systems

LPF - Low Pass Filters

ADC - Analog to Digital Converter

TTS - Text to Speech SPL - Sound Pressure Levels

SVM - Support Vector Machine

SR - Speech Recognition

AM - Amplitude Modulation

AIC - ctive Inaudible-Voice-Command Cancellation

ANC - Active Noise Cancellation

STFT - Short-Time Fourier Transform

RA - Recognition Accuracy

EER - Equal Error Rate

FAR - False Acceptance Rate

FRR - False Rejection Rate

SNR - Signal to Noise Ratio

VAD - Voice Activity Detection

TDoA - Time Difference of Arrival

DPAM - Deep Perceptual Audio Metric

PSTN - Public Switched Telephone Network

LTAS - Long-Time Average Spectrum

VAE - Variational Autoencoder

AUC - Area Under Curve

CNN - Convolutional Neural Network

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Chapter 1

Introduction

1.1 Preamble

The use of speech recognition to interact with devices is becoming more and more popular. Voice assistants are being used for a wide range of purposes such as ordering for food or calling a friend. Apart from the benefits that these systems offer, it was discovered that these systems respond to malicious and sneaky attacks that are unrecognizable to humans, but not to these systems. For example, an attacker can send a physically inaudible voice command to a smart speaker and make it open the home door without being heard by the user.

Humans can hear sounds in the range between the frequencies of 20Hz and 20,000 Hz. Any sounds outside of this range are inaudible to people. Inaudible Voice Attacks involve sending malicious commands at frequencies beyond the range of human hearing. These attacks manipulate devices that are programmed to recognize voice commands without the user being aware. By embedding voice commands into ultrasonic signals, attackers can silently control these devices from a distance.

A group of researchers demonstrated this attack first in 2017, which they called DolphinAttack. They were able to initiate a FaceTime call on iPhone, activate Google Now to switch the phone to airplane mode and even manipulate the navigation system in an Audi automobile without being heard by a person [1]. This class of attacks poses a significant threat as it bypasses traditional auditory defenses, allowing malicious actors to trigger unintended actions on victim devices without alerting the user. With the growing reliance on voice assistants in everyday tasks, the need for defenses against inaudible voice attacks has become critical.

1.2 Scope and Motivation

To mitigate such threats, existing studies have suggested several hardware-based and software-based countermeasures. Though hardware redesign can fix the vulnerability for future devices, smartphone companies indicate that software mitigation that can instantly protect various existing devices without any hardware modifications are still in desperate need. Existing devices, for example, smartphones, wearables, and smart speakers, are designed with diverse microphone hardware, voice assistant software, and computational resources, which require the defense to be transferable and lightweight. Research discovered that patterns of inaudible voice attacks are diverse whereas benign audios share similar patterns. This observation was one of the reasons that served as a motivation for the development of a method in detecting attacks (the anomaly) only by learning benign audios (the normality).

1.3 Objectives

The seminar aims to design a holistic software-based mitigation that can detect inaudible voice attack for many smart devices only based on accessible normal audios. The specific goals of the design are:

- Lightweight: On considering resource limitation in IoT devices, the algorithm needs to be lightweight and requires minimal computational overhead and storage resources.

- Universal: The defense is expected to be instantly applied to a wide range of devices without requiring any hardware modification.

- Unsupervised: Training an unsupervised system does not require inaudible data, so as to let researchers and manufacturers free from expensive equipment and costly data acquisition.

1.4 Organization of the Report

The report is organized as follows:

- Area of Research, which explains the current state of art, the issues and threats associated with the problem at hand and potential applications of the methods developed.
- Methods, which details down the methodologies proposed by each of the literature papers that was surveyed to carry out the seminar.
- Conclusion, which concludes the report by performing a comparison of the various methods used by literature, including it's advantages and disadvantages.

1.5 Summary of the Chapter

This chapter provides an introduction to Inaudible Voice Attacks, the significance of defending against such an attack, the scope & motivation for the seminar and the objectives of the study carried out on Inaudible Voice Attacks.

Chapter 2

Area of the Research

This chapter discusses the current state of art in the area under research on defending against inaudible voice attacks, the issues and threats that these attacks pose for various voice controlled systems and the applications of developing defenses for the same.

2.1 Current State of Art

Voice-controlled systems (VCS) have experienced rapid advancements due to the integration of speech recognition (SR) technologies in everyday devices such as smartphones, smart speakers, and Internet of Things (IoT) applications. Popular systems like Siri, Google Now, and Alexa have become essential to user interactions, enhancing convenience and improving user experiences through voice commands. However, this widespread use has also highlighted significant security risks, especially in relation to inaudible voice commands.

Recent research has focused on improving the resilience of SR systems against various attack methods. Ultrasonic attacks, which exploit frequencies above 20 kHz (beyond the range of human hearing), have been identified as a serious threat. Studies have shown that SR systems could unintentionally process ultrasonic signals, resulting in unauthorized commands being executed. Moreover, innovations in signal modulation techniques have made it easier to embed malicious commands within otherwise harmless audio streams, complicating detection efforts.

Most existing defense mechanisms are based on hardware solutions, such as the use of high-pass filters to block ultrasonic frequencies, and software approaches like anomaly detection algorithms that flag irregular audio patterns. There are also a few software-based defenses that use machine learning models, including Convolutional Neural Networks (CNN), Logistic Regression, Recurrent Neural Networks (RNN), Variational Autoencoders (VAE),

and Support Vector Machines (SVM), to classify and detect malicious audio patterns. Despite these efforts, there are still gaps in effectively addressing nonlinear distortions caused by microphone and amplifier hardware, which can inadvertently demodulate ultrasonic signals into audible commands.

2.2 Issues and Threats

The issues and threats that are associated with Inaudible Voice Attacks are:

1. Unauthorized device control : Once an attacker gains access to the device via DolphinAttack, they can perform actions ranging from privacy invasions (e.g., listening to conversations) to manipulating critical systems in smart homes or offices. For example, a hacker issues an ultrasonic command to "Alexa" to disable a home security system, gaining access to the house.

2. Data theft : Voice-activated devices often have access to personal information, such as emails, calendars, contact lists, and other sensitive data. DolphinAttack can inject inaudible commands to manipulate these devices into extracting or sending this data to the attacker. For example, a voice command telling Google Assistant to read out the latest banking notification, revealing sensitive financial details to the attacker.

3. Financial Manipulation : Many smart assistants and voice-activated devices are integrated with financial services. DolphinAttack can exploit these information to conduct unauthorized transactions or manipulate financial data. For example, an ultrasonic command prompts the assistant to "order" a high-priced item from an online store, charging the user's account without their knowledge.

4. Physical damage : In modern smart homes and IoT environments, many physical systems (such as door locks, appliances, lighting, or even vehicles) are controlled by voice commands. DolphinAttack can manipulate these systems, causing physical harm or damage. For example, a hacker could use ultrasonic commands to manipulate an autonomous vehicle's navigation system, sending incorrect signals that

cause it to veer off course or disabling the vehicle's safety protocols through ultrasonic commands, preventing it from auto-correcting or applying emergency braking in dangerous situations.

2.3 Potential Applications

Detection techniques for inaudible voice attacks have a wide range of potential applications, particularly in security-sensitive environments where voice-controlled devices and systems are used. These detection methods can be applied in various industries to enhance device safety, prevent unauthorized actions, and protect sensitive data.

1. Voice-Activated Devices (Smart Assistants and IoT)
    - Smart Speakers (e.g., Amazon Echo, Google Home): Voice assistants are increasingly common in homes and offices. Detecting inaudible voice commands can prevent unauthorized manipulation of these devices, such as changing settings, activating devices, or making purchases through voice commands.
    - Home Automation: Many home automation systems, including lighting, security, and temperature control, are voice-activated. Protecting these systems from ultrasonic command injection ensures that only authorized users can make changes.
2. Automotive Systems
    - Voice-Activated In-Car Systems: Modern vehicles often come with voice-activated controls for navigation, entertainment, and climate. Preventing ultrasonic attacks is essential for protecting against unauthorized actions, such as changing navigation routes, unlocking doors, or even disabling critical safety features.
    - Driver Authentication: Some cars use voice recognition for driver authentication. Detecting ultrasonic signals can prevent impersonators from gaining access to the vehicle or tampering with sensitive features.
3. Government and Military Applications

- Secure Communication Systems: In sensitive government or military environments, detecting ultrasonic voice commands is essential for preventing espionage, unauthorized control of communication systems, and command-and-control systems that could lead to security breaches.
- Voice-Activated Security Systems: Government agencies and military installations often use voice-controlled security and access control systems. Preventing ultrasonic voice commands from bypassing these systems is crucial to maintaining national security and protecting sensitive information.

2.4 Summary of the Chapter

This chapter highlights the current state of state for defending against inaudible voice commands, the issues and threats associated with such an attack and the potential applications of the defenses in securing systems across various fields.

Chapter 3

Methods

This chapter aims to provide a detailed explanation on the various methodologies developed to prevent and detect attacks involving inaudible voice commands.

3.1 DolphinAttack : Inaudible Voice Commands (2017)

This paper presents a security vulnerability in voice-controllable systems (VCS), where inaudible ultrasonic commands can be exploited to manipulate devices without users noticing. The authors show that these "hidden" commands can initiate malicious actions, like visiting harmful websites, making unauthorized calls, or disabling wireless communication. Through successful testing on systems such as Siri, Alexa, and Google Now across multiple devices, the study underscores the need for improved security in voice recognition technology and suggests hardware and software defenses to mitigate such attacks.

3.1.1 Background Model

Voice Controllable System

Voice-controllable systems (VCS) typically consist of three components:

1. Voice Capture
2. Speech Recognition
3. Command Execution

The voice capture subsystem records sounds, filtering and amplifying signals, which are then processed in two phases. First, an activation phase, triggered by a wake word or key, initializes recording. Once activated, the recognition phase converts spoken commands into text, usually using cloud-based, machine-learning algorithms. Commands are then

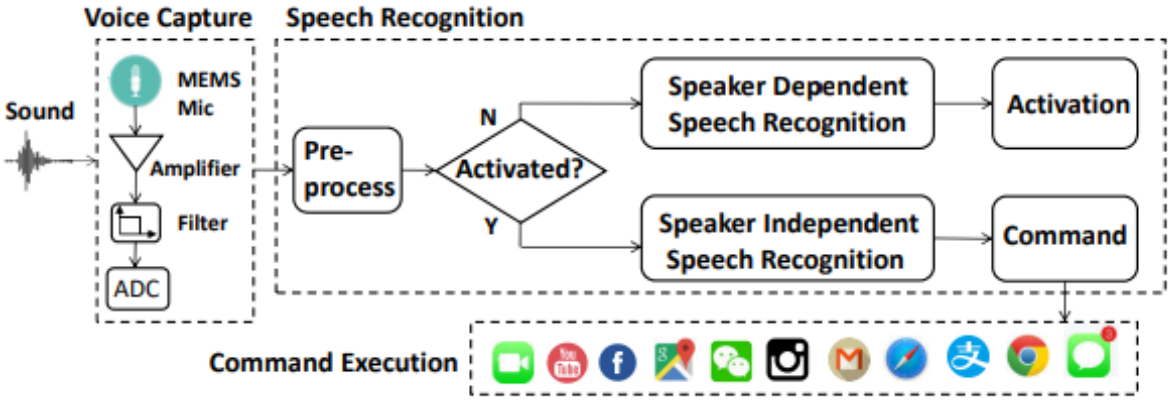


Figure 3.1: Architecture of Voice Controllable Systems

executed by system-defined actions. These VCS systems are vulnerable to inaudible ultrasonic attacks.

Microphone

The voice capture subsystem in voice-controlled devices uses microphones, especially Micro Electro Mechanical Systems (MEMS) microphones, which convert sound waves into electrical signals. These microphones, alongside low-pass filters (LPFs) and analog-to-digital converters (ADC), are designed to capture sounds within the human hearing range (20 Hz to 20 kHz). Signals above this range, such as ultrasonic frequencies, are typically filtered out, though vulnerabilities still exist for inaudible attacks.

Threat Model

The adversary's objective is to covertly inject unauthenticated voice commands into voice-controlled systems without user awareness. They assume no physical access to the device, no owner interaction, and rely solely on ultrasonic (inaudible) commands above 20 kHz to avoid detection. Attackers can use equipment such as ultrasonic-capable speakers placed near the target device, allowing commands to be executed remotely and discreetly.

3.1.2 Nonlinearity

The DolphinAttack concept relies on modulating low-frequency voice commands onto an ultrasonic carrier and allowing the device's non-linear microphone components to demodulate the signal. Because microphones and amplifiers introduce non-linear effects, they

can create new frequencies when ultrasonic signals are introduced. By using amplitude modulation and carefully selecting frequencies, DolphinAttack downconverts the ultrasonic command signal, which remains after the low-pass filter, making it interpretable by voice-controlled systems while staying inaudible to humans.

3.1.3 Attack Design

DolphinAttack enables covert control of voice-controlled systems (VCSs) by injecting inaudible voice commands via ultrasonic modulation. To activate a device like Siri, it generates activation commands like "Hey Siri" through Text-to-Speech (TTS) brute force or by concatenating phonemes from recorded speech. This had a success rate of 39% in testing. Control commands like "call 911" follow activation without authentication checks, so they are injected more easily. Two transmitter types facilitate the attack: a powerful lab-based setup for robust testing and a portable smartphone-based design to expand attack feasibility in real-world scenarios. The portable setup, using a smartphone and ultrasonic transducer, has limited range than the powerful one, but demonstrates the attack's adaptability for everyday devices. Key parameters like modulation frequency and depth are carefully selected to ensure inaudibility and effective VCS response. This attack model shows the potential for widespread VCS vulnerability to silent, remote control.

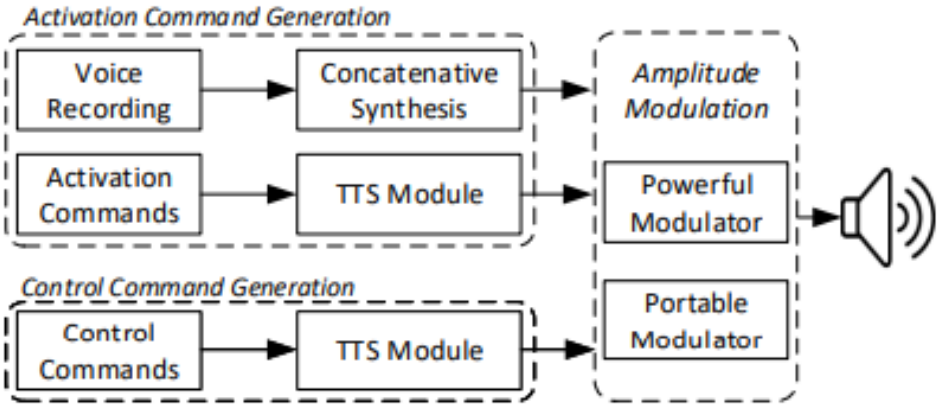


Figure 3.2: Design of DolphinAttack

3.1.4 Impact Quantification

DolphinAttack's effectiveness varies across languages, background noise levels, sound pressure levels (SPLs), and distances. Tested commands in five languages showed high recog-

nition rates, with English and Spanish achieving 100%. Noise affects control commands more than activation commands, with recognition rates dropping in louder settings. SPL significantly impacts success, as higher SPLs improve signal-to-noise ratios and recognition accuracy, especially for the Apple Watch. Attack range also varies, with Apple devices responding from further distances than Samsung devices. Portable device attacks succeeded at close range but were limited by speaker frequency constraints.

3.1.5 Defenses

To counter DolphinAttack, two defense strategies are proposed:

1. **Hardware-Based Defense:** This involves enhancing microphones to block ultrasonic signals above 20 kHz and adding modules to detect and cancel inaudible voice commands before they reach the device.

2. **Software-Based Defense:** Using machine learning classifiers, like SVM, to distinguish between legitimate and malicious modulated voice commands. This approach detects unique frequency characteristics (500-1000 Hz) present in modulated attacks. Testing showed a 100% accuracy rate in distinguishing attack signals from genuine ones.

These defenses together aim to prevent unauthorized, inaudible command injections.

3.1.6 Conclusion

The paper introduces "DolphinAttack," an inaudible attack on speech recognition (SR) systems like Siri, Google Now, and Alexa. Using amplitude modulation (AM), it embeds voice commands in ultrasonic frequencies, making them undetectable by humans. To counter potential misuse, the authors propose two defense methods, addressing both hardware and software aspects to protect SR systems from these stealthy attacks. [1]

3.2 Canceling Inaudible Voice Commands Against Voice Control Systems (2019)

This paper fills the gap created by the lack of a complete defense mechanism to be able to not only detect the attack but also cancel out the attack signal, and meanwhile ensure the

legitimate voice commands unaffected. The methodology proposed by this paper is called Active Inaudible-Voice-Command Cancellation (AIC) system, which involves detecting and canceling inaudible attacks by creating an "anti-signal" based on the unique spectral pattern that the attack signal produces when it interacts with a continuous guard signal. This process relies on the non-linear behavior of microphones, which causes the ultrasonic attack and guard signals to produce detectable byproducts in the VCS microphone's passband.

3.2.1 Motivation

This method is inspired by the active noise cancellation (ANC) technique used in devices like headphones, which cancels the environmental noise by producing antinoise signals close to human ears. The architecture of ANC is shown in Fig.3.3.

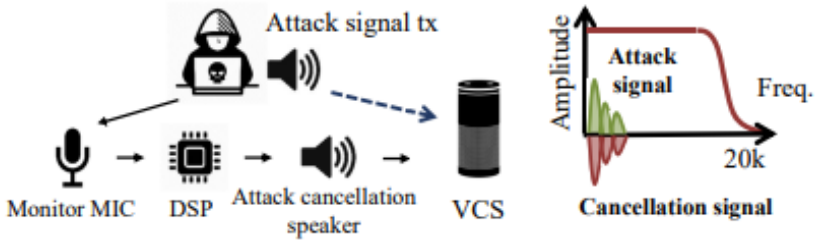


Figure 3.3: Motivation : ANC architecture

A monitor microphone is responsible to detect and capture the attack signal. The DSP unit computes an anti-attack signal that is factually reverse to the attack signal, which is then sent out by the attack cancellation speaker to neutralize the inaudible voice command.

3.2.2 Methodology

The designed guarding scheme involves utilizing multi-tones used to generate two copies of the attack signal into the mic's passband. The multi-tones are a set of ultrasonic signals with frequency interval of 20kHz. The VCS's mic passband is divided into 2 sub-bands as shown in Fig.3.4 , where the signal in the left sub-band is s(t) and that in the right subband is r(t).

The basic idea of signal cancellation is to create an exact copy of the target signal

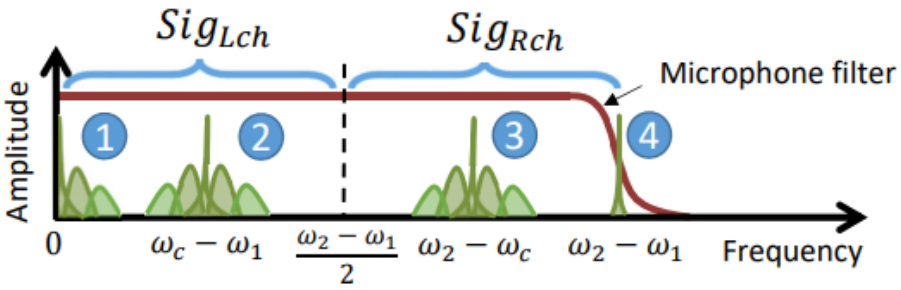


Figure 3.4: Spectrum by Guard signals

and use the copy to "subtract" the target, where the copy creation can be realized by designing an adaptive filter.

The working of the AIC is as follows:

1. Guard Signal Transmission

   The AIC system broadcasts ultrasonic guard signals at specific frequency intervals (20kHz apart) around the attack frequency range. These guard signals are inaudible to humans and do not interfere with regular voice commands but set a baseline that the microphone can detect if an attack signal is present.

2. Signal Interaction and Detection

   When an attack signal interacts with the guard signal, the non-linearity of the microphone causes the signals to combine and create distinct frequency components in the microphone's audible range. These new frequency components serve as a "fingerprint" of the attack, making it clear when an inaudible attack is occurring.

3. Sub-band Analysis and Anti-signal Creation

   The AIC system divides the microphone's passband into two sub-bands—one for the attack signal's representation and one for the guard signal's response. By analyzing the frequencies and patterns in these sub-bands using a Short-Time Fourier Transform (STFT), the system identifies the presence and characteristics of the attack signal.

4. Generating the Anti-signal

Once the attack signal's signature is detected in one sub-band, the AIC system creates an "anti-signal" that mirrors this signature. This anti-signal is a phase-inverted version of the attack signal, designed to cancel it out when the two signals interact. Since the signals are now out of phase, when they combine, they effectively "subtract" from each other, neutralizing the original attack signal.

5. Signal Cancellation

This anti-signal is applied within the passband of the microphone. When the inverted attack signal combines with the original attack signal, they cancel each other out, eliminating the impact of the attack without disrupting the microphone's ability to pick up legitimate voice commands. This allows the VCS to continue functioning as normal, only recording and processing commands that do not match the attack pattern.

Two versions of the guard signal generator were implemented for different scenarios. The small one is made of 8 small speakers and two AD9833 chips to realize short range or mobile protection. The big one is made of 18 speakers, two AD9833 chips and power amplifier to realize long range protection.

3.2.3 Performance

Without the legitimate voice commands, our system can protect the VCS from the inaudible attack in 100% if the attack signal's frequency is not in the middle point of the neighboring guard signals, otherwise the success rate is about 99%. Legitimate voice commands still can be recognized in 98% even if it comes with the attack signal simultaneously, with no impact on the normal VCS use. [2]

3.3 Robust Detection of Machine-induced Audio Attacks (2021)

Existing defense strategies have focused only on single and dual-channel audio, while very few papers have worked on multi-channel microphone array for defending against specific types of attacks by treating each attack differently. The paper is driven by the motivation of research on defending miscellaneous audio attacks, and the potential benefits of multi-

channel audio. Compared to existing single-channel-based solutions,the benefits offered by multi-channel audio attack detection are :

1. Spatial Feature
2. Detection range
3. Device compatibility

The overview of the proposed approach is given in Fig.3.5

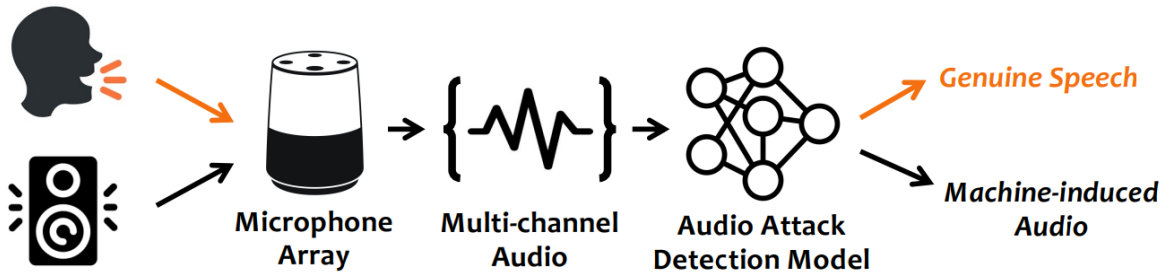


Figure 3.5: Proposed Approach

3.3.1 Methodology

This paper builds a holistic solution for detecting machine-induced audio attacks using multi-channel microphone arrays already available on modern intelligent audio systems. It utilizes magnitude and phase spectrograms of multi-channel audio to extract spatial information which is used to train a deep learning model to detect the fundamental difference between human speech and adversarial audio generated by the playback machines. The method is inspired by the fact that all audio attacks require a playback device (e.g., loudspeaker or ultrasound speaker) to play the crafted attack speech sample, while genuine speech is uttered from human vocal cords. This inherent difference of sound production will be carried over into the produced audio, resulting in different patterns in signal frequency and directivity , which can be captured by microphone arrays and further utilized to differentiate genuine speech from machine-induced attack audio. Making use of both spatial and temporal information from multi-channel audio, the system can be trained to detect various types of voice attacks. It also adopts an unsupervised domain adapta-

tion training framework to further improve the model's generalizability in new acoustic environments.

The design objectives are that :

1. the model should be able to utilize the rich information encoded in multi-channel audio to achieve enhanced audio attack detection accuracy compared to existing single-channel based methods

2. in order to build a holistic defense against any machine-induced audio attack, the model should be able to capture a set of generic acoustic features that distinguish genuine speech from machine-induced audio

3. the model should rely on environment-independent features only to maintain a decent detection performance in different acoustic environments

Explicitly collecting labeled data for all common acoustic environments is difficult in practice and thus it will be more desirable to enable the model to generalize to new acoustic environments without requiring labeled data for achieving robust defense. The detection model relies on a short segment of the audio recorded by the microphone array to make a decision. The length of the segment should be set as short as possible to achieve timely detection for real-time applications while maintaining a high recognition accuracy. The input given to the network of the model is a unified time-frequency map produced by applying the Short-Time Fourier Transform (STFT) on the audio signal.

Two types of network configuration are used according to the implementation scenario.

1. Type I : a large and powerful network that has more representational capacity to enable high attack detection accuracy for desktop or cloud. It is is built upon the VGG-16 network, with the number of input channels modified according to the multi-channel audio and the number of output neurons set to 2.

2. Type II : fast and lightweight network that provides more computational and energy savings, which makes it suitable for mobile and IoT deployments. It is based on the MobileNetV2, that accommodates the multi-channel audio attack detection task.

The overall structure of the proposed audio attack detection network is presented in Fig.3.6. The network consists of 3 components:

1. a CNN feature extractor

2. a fully-connected (FC) genuine/attack audio classifier

3. an optional domain discriminator which is only involved in environment independent training

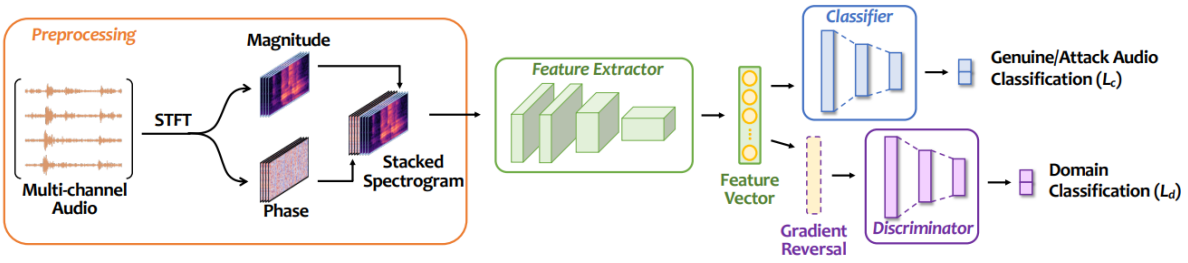


Figure 3.6: Proposed audio attack detection network

As the model relies on the training data heavily to learn to make predictions, if the training samples do not accurately model the distribution of the actual data, the performance will likely degrade. This poses a challenge as in practice, the model will definitely face new acoustic environments that was not represented in the data during training. Hence, a domain adaptation process is employed to help the model generalize from the training-time environment (i.e., the source domain) to the test-time environment (i.e., the target domain). Consider w_f and w_c to be the parameters of the feature extractors and classifier, respectively. The network is trained on classification loss L_c(w_f, w_c). A domain discriminator with parameter w_d is introduced during training so that the feature extractor learn domain-invariant features. The domain discriminator has the same architecture as that of the classifier but the objective of the discriminator is to distinguish between the source domain training samples and the target domain training samples by minimizing the domain classification loss L_d(w_f, w_d). The objective of the domain-invariant training process is to search for the parameter set w_f to minimize the audio classification loss L_c and simultaneously maximize the domain classification loss L_d. After training, the feature extractor will learn to extract features that are both discriminative for detecting various audio attacks and invariant to the change of acoustic environments.

3.3.2 Performance Evaluation

Evaluation is conducted under various settings on a public multi-channel replay attack dataset and a self-collected multi-channel audio attack dataset involving 5 types of advanced audio attacks. For advanced attacks, such as inaudible voice attacks,the paper uses a self-collected dataset, where 3 representative microphone arrays were used to record the attack audio and genuine human speech.

2 evaluation metrics have been used to evaluate the performance of the system, which are:

1. Recognition Accuracy (RA): Audio attack detection can be viewed as a binary classification problem. Recognition accuracy is the percentage of audio samples being correctly classified.

2. Equal Error Rate (EER): EER is a commonly used metric for evaluating replay attack detection system. It depends on two detection error rates: the false acceptance rate (FAR) and the false rejection rate (FRR). EER corresponds to the point at which the two detection error rates are approximately equal.

According to the results, the Type I network was able to achieve an overall average EER of 12.9% across all 3 devices used for testing. The proposed Type II network was also able to achieve a relatively high recognition accuracy and low EER that surpasses existing single- and multi-channel-based models in most cases. [3]

3.4 EarArray (2021)

This paper designs EarArray, a lightweight method that can not only detect such attacks but also identify the direction of attackers without requiring any extra hardware or hardware modification.

Instead of relying on signal distortion from the microphones, EarArray focuses on the differences in how inaudible (ultrasonic) and audible voice commands propagate. As shown in Figure 3.7, when a command is issued from a speaker, the sound travels to the smart device (e.g., a smartphone or smart speaker) and reaches its microphones at different times based on the distances between the speaker and each microphone. Each

microphone on the device receives the sound with an amplitude that decreases proportionally to the square of the distance and the attenuation rate. This attenuation rate is proportional to the square of the sound frequency. Thus, by using measurements from multiple microphones on the device (e.g., a microphone array), EarArray can determine whether the command is audible or inaudible, since ultrasound frequencies (above 20kHz) decay faster than audible frequencies (typically below 5kHz for human voice).

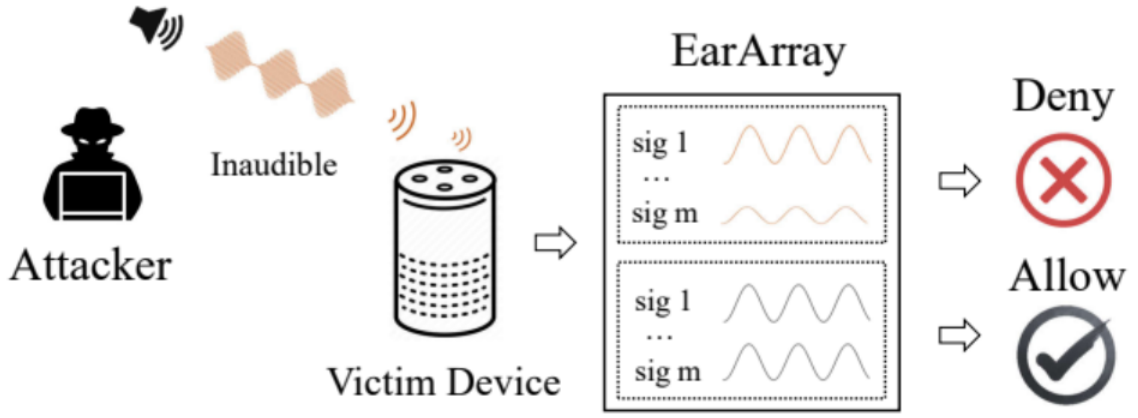


Figure 3.7: EarArray

EarArray relies on the physics of sound propagation, and will not be affected by the microphones' hardware characteristics.

3.4.1 Methodology

The methodology used in the EarArray system for detecting inaudible voice command attacks consists of four main stages: audio signal preprocessing, feature extraction, attack detection, and localization, as shown in Fig.3.8.

1. Audio Signal Preprocessing This stage processes the raw audio input to remove noise and non-speech components, preparing it for further analysis.

- De-noising: Due to environmental noise, audio signals captured by microphones are often noisy. To improve the signal-to-noise ratio (SNR), a band-pass filter is applied. The filter allows only frequencies within the human speech range (50 Hz to 2 kHz) to pass through, eliminating unwanted frequencies outside this range.

- Voice Activity Detection (VAD): The system identifies parts of the audio that contain speech and removes non-speech intervals. It uses a VAD algorithm that detects

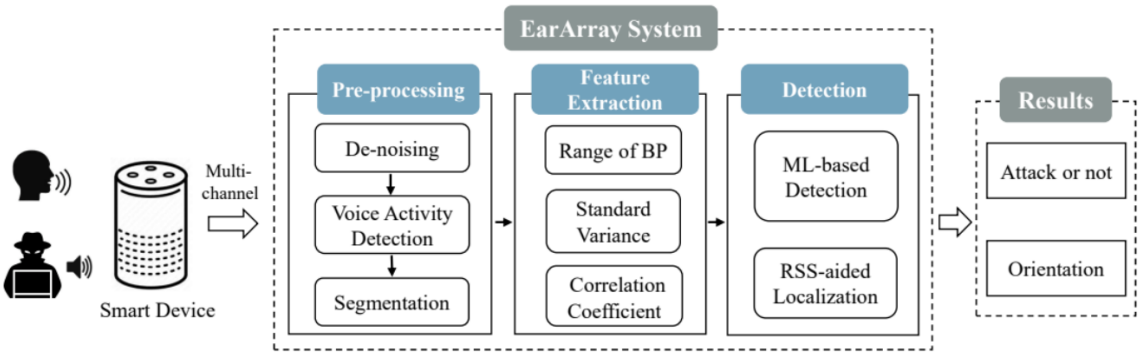


Figure 3.8: EarArray System

the presence or absence of speech based on band power (the power within a specific frequency range).

- Segmentation: The filtered signal is segmented into chunks with a specific duration to create manageable portions for analysis. Any segment with band power below a set threshold is discarded, as it likely does not contain speech. This ensures that only relevant speech-related data moves on to the next stage.

2. Feature Extraction In this stage, relevant features are extracted from the preprocessed audio segments to characterize the sound's spatial properties. These features are designed to differentiate between normal human voice commands and inaudible attacks.

- Range and Standard Deviation of Band Power: For each audio channel, the system calculates the range and standard deviation of the band power, which reflects the intensity of the sound. Since human speech occupies a narrow frequency range, these metrics help to differentiate it from attack signals.

- Pearson Correlation Coefficient: The correlation coefficient is used to measure the similarity between the signals across different channels. Since inaudible attack signals usually have a more uniform spatial distribution compared to human speech, the correlation values can help distinguish between the two. The system computes the correlation in the frequency domain to eliminate the impact of phase differences, focusing solely on energy distribution patterns.

3. Attack Detection This stage uses the extracted features to identify inaudible voice command attacks.

- Machine Learning-Based Classification: EarArray employs a SVM classifier, trained on both normal and inaudible attack samples collected in different scenarios.
- Detection Phase: In real-time operation, EarArray calculates the features for incoming audio samples and feeds them to the trained SVM model. The model then classifies each sample as either a normal voice command or an inaudible attack. This classification allows the system to detect and flag potential attacks.

3.4.2 Attack Localization

Once an inaudible attack is detected, EarArray attempts to determine the direction of the attacker.
Conventional localization techniques, such as TDoA, are less effective for inaudible commands due to high-frequency attenuation, which weakens the signal strength on some channels. This makes it hard to accurately pinpoint the direction of the sound source. To overcome this challenge, EarArray uses a band-power-based method. It calculates the band power for each microphone channel to estimate the relative strength of the signal across channels. The microphone with the highest power reading indicates the general direction of the attack source. This method provides a coarse direction rather than a precise location, allowing users to get a general sense of the attacker's position.

3.4.3 Performance

EarArray can detect the inaudible voice commands with accuracy of above 99%, meanwhile, the accuracy of localization is up to 97.89%. [4]

3.5 Learning Normality is Enough (2023)

To protect the existing and future devices from the threats caused by inaudible voice attacks, the paper proposes NormDetect, a software-based mitigation that can be instantly applied to a wide range of devices without requiring any hardware modification. Unlike the approaches devised by existing studies which are supervised in nature, this paper adopts an unsupervised learning method to detect such attacks. It involves training the model on benign audio samples rather than on both kinds of samples.

3.5.1 Attack Investigation

Before designing the defense, several experiments were conducted to investigate audible and inaudible voice attacks. The investigation was driven by 3 research questions:

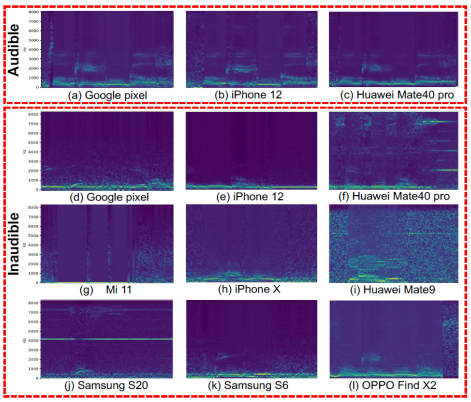
- RQ 1: What are the differences between benign audio and inaudible voice attacks?
- RQ 2: How different are the benign audio and inaudible voice attacks?
- RQ 3: Why do the attack patterns vary on different target devices?

RQ1 : What are the differences between benign audio and inaudible voice attacks? To answer RQ1, the audible benign voices and inaudible voice attacks were recorded on 3 commercial smartphones: iPhone12, Huawei Mate40 pro, and OPPO Find X2 with variations in distances (from 10 to 300 cm), attack powers (from 0.1 to 1 Watt), voice commands, and target devices. The observations are:

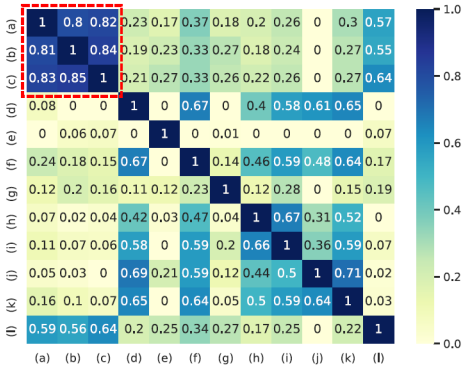
- The quality of the injected attack audio varies greatly with the target device. Experimental results showed that the recorded attack commands sound clear on Find X2, relatively clear on iPhone12, and slightly clear on Mate40 pro due to obvious noise.
- The quality of the recorded benign voices is clear and similar across all devices.
- The attack power directly affects the audio quality. During the experiment, the attack audio on Mate40 pro sounded like a scream when the attack power was exceeded beyond a certain threshold.
- Speech commands have little impact on the attack effectiveness. The experiment used 5 TTS commands as the baseband, and the qualities of the injected attack audio are almost the same.

RQ2 : How different are the benign audio and inaudible voice attacks? To answer the RQ2. an experiment was conducted to quantify the difference between normal and inaudible voice commands received by 24 smart devices. The subfigures (a,b,c) in Fig.3.9(a) demonstrate an audible command "OK Google" recorded by 3 smartphones and subfigures (d-l) indicate the inaudible voice attack with the same command recorded

by 9 different smartphones. It can be observed that the spectrums of the 3 audible cases are quite similar whereas the patterns of (d-l) are different from each other. To precisely quantify the similarities and differences, a Deep Perceptual Audio Metric is employed. The DPAM is a metric designed to evaluate how perceptually similar two audio signals are to a human listener. The output of a deep perceptual audio metric is a numerical score that indicates the perceptual similarity between two audio signals. The range of values lies between 0 and 1, where 0 indicates complete dissimilarity and 1 indicates complete similarity. On observing the matrix in Fig.3.9(b), it can be inferred that the audible commands have close similarity, whereas the inaudible commands have lesser similarity score. From this it was concluded that benign audios share similarities, while inaudible audios do not.



(a) STFT Spectrums of audible and inaudible commands



(b) Similarity Matrix computed using DPAM

Figure 3.9: Spectrums of commands and similarity matrix

RQ3 : Why do the attack patterns vary on different target devices? The main reason why inaudible voice attack works, is due to the non linearity feature of the microphone, hence microphones may be a dominant factor in understanding the diverse patterns of inaudible commands. The paper surveyed 10 microphones and the frequency responses of 6 of them are shown in Fig.3.10. It is found that the microphones share a similar frequency response characteristic within the audible frequency bands. The microphones have been designed to comply to the PSTN (Public Switched Telephone Network) standard which restricts the sound waves to be within the range of 300Hz to 3400Hz. So the frequency response of different microphones in this band is calibrated to be almost

identical, which explains why the spectrums of normal voices are so similar. In the bands over 10 kHz, the response of different microphones is distinct. Hence, it can be concluded that the root cause for the different attack patterns on various devices is the diverse frequency response of microphones in the upper 10 kHz band.

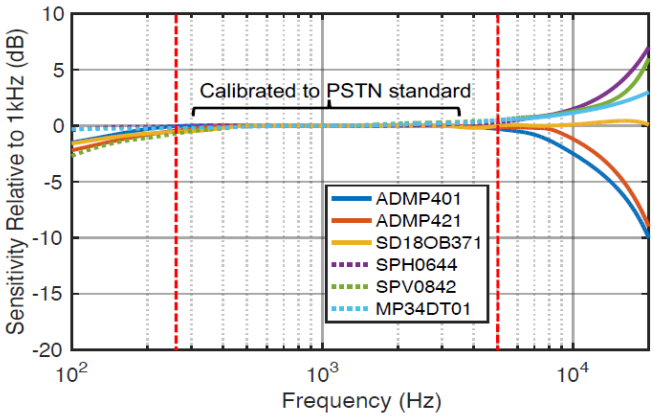


Figure 3.10: Frequency response curves

3.5.2 System Overview

The overview of the system proposed by the paper is as shown in Fig.3.11.
The key stages of the system are:

1. Speech Preprocessing
2. Spectrum Augmentation
3. NormDetect Model.

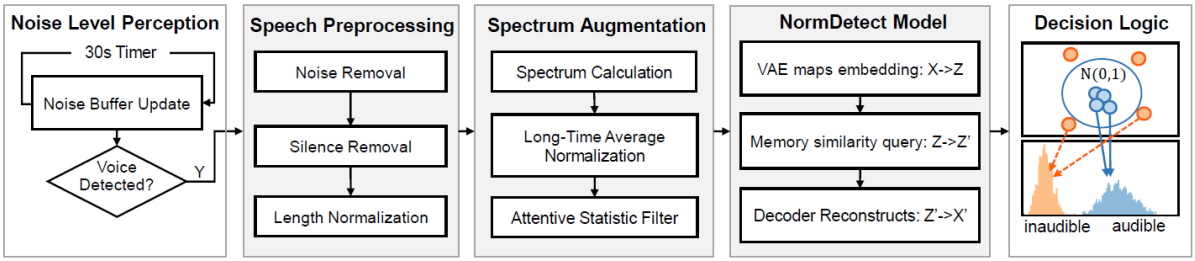


Figure 3.11: System Overview

Prior to the key stages, the system consists of a module called Noise Level Perception, which is a method that makes the system continuously monitor the ambient noise in

its environment. The system maintains a buffer of the last five noise samples, which it continuously updates. Each time a new noise sample is recorded, it enters this queue, replacing the oldest one. The system calculates the average noise level from these five samples, to obtain a more stable estimate of the overall ambient noise level. The noise buffer is updated periodically by a timer to account for changes in the noise environment. Once the system has a stable understanding of the ambient noise, it can effectively detect any new, distinct sound, such as a voice command, by comparing the new sound against the calculated noise level estimate. If a voice is detected, the system moves to the next stage.

1. Speech Preprocessing

1. Noise Removal : Ambient noise is found to have an abnormal response similar to that of inaudible commands. Hence it is essential to remove ambient noise to isolate the response of inaudible commands.

2. Silence Removal : It is important to eliminate the influence of different unvoiced segments, which are caused by speakers' habits, such as speaking speed, and semantic pauses. For example, when people say "OK Google", the pause duration and speech speed vary among young and old people. Such pauses must be eliminated as much as possible to ensure that the model can focus on learning voiced features.

3. Length Normalization : It is performed to ensure that all voice commands, regardless of their original duration, have a consistent input length when fed into the machine learning model.

2. Spectrum Augmentation

This module transforms normalized audio into augmented spectrums.

1. Spectrum Calculation/Generation : In this step, feature extraction is performed using the Short-Term Fourier Transform, which is a frequency domain feature. The STFT converts an audio signal into its frequency components over time, allowing the system to analyze how the frequency content of the signal changes throughout the duration of the voice command, which is critical for identifying and differentiating between voice commands and possible inaudible attacks.

2. Long-Time Average Normalization : This method generates a Long-Time Average Spectrum (LTAS), by averaging the log STFT spectrums over time, which minimizes variations caused by different speech content and speaker differences. This helps ensure that the system can focus more on the underlying patterns by reducing the impact of irrelevant speech variations.

3. Attentive Statistical Filter : The Attentive Statistical Filter is used to enhance the discriminative information between audible and inaudible voice commands by suppressing irrelevant speaker-specific and linguistic variations, allowing the system to focus on features that differentiate the two classes more effectively. For this, the F-ratio method is used. The F-ratio method is applied to the Long-Time Average Spectrums (LTAS) of normal audio samples. which helps identify which sub-bands contain the most significant distinguishing information between normal and inaudible audio. A 128-dimensional weight coefficient vector is used as a filter to each input LTAS, assigning higher weights to more significant sub-bands (where differences between audible and inaudible commands are more significant) while reducing the weights for less significant sub-bands.

3. NormDetect Model

The main principle behind the model is that the model reconstructs the normal data well based on standard training data, while it reconstructs poorly for abnormal data. The model is a Variational Autoencoder (VAE), to which the augmented spectrum is fed as input. The VAE then performs mapping, reconstruction, and judgement based on the anomaly score. The diagram of the proposed model consisting of 3 stages is shown in Fig.3.12.

STAGE 1: VAE maps embedding $X \rightarrow Z$

In this stage, the encoder produces a latent embedding Z of the input spectrum X. The embedding Z follows a Gaussian distribution and captures the essential features of the data.

STAGE 2: Memory Similarity Query $Z \rightarrow Z'$

This stage involves a memory network structure, also called the memory module, that

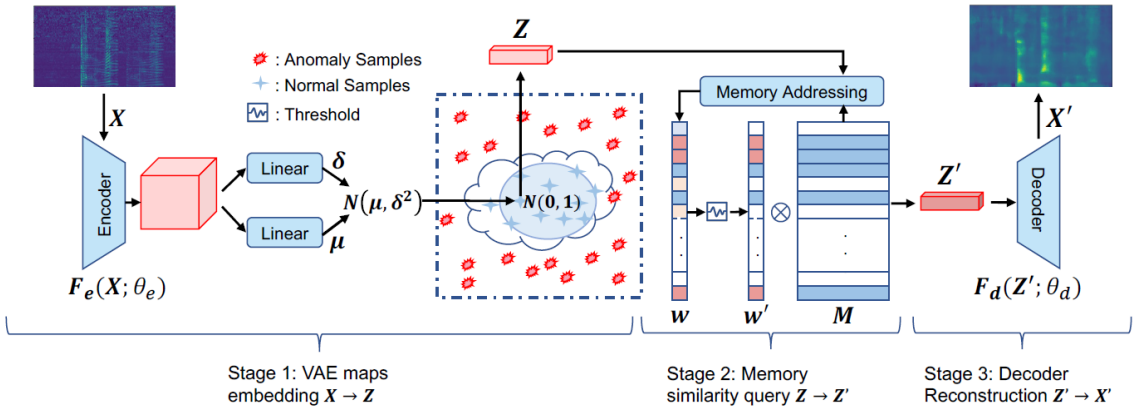


Figure 3.12: NormDetect Model

consists of multiple items M_i that represent prototypical patterns of normal audible voice commands. The mapped embedding Z is compared to each memory item M_i to compute a similarity score. This generates a similarity weights vector w. Low-similarity items in w are removed, resulting in a refined weights vector w'. The refined weights vector w' is then used to compute Z' based on the most similar memory items, which reflects the closest match to the normal voice command patterns, helping to mitigate the risk of false acceptance by voice command systems.

STAGE 3: Decoder Reconstruct $Z' \rightarrow X'$

The decoder takes the embedding Z' to reconstruct the input X'. An anomaly score is computed between X and X' by taking the negative log-likelihood.

3.5.3 Conclusion

The NormDetect model was evaluated on a dataset containing 383320 samples which included both audible and inaudible voice commands. The results suggested that the model demonstrates strong overall discriminatory power, real-world applicability and adaptibility across various devices as it performed an average AUC of 99.48% and EER of 2.23%. [5]

3.6 Summary of the Chapter

This chapter explores the various methods that have been developed to detect DolphinAttack, focusing on properties of ultrasonic signals to secure voice-controlled systems. EarArray utilizes a microphone array to capture differences in signal propagation and attenuation rates, using features like band power variance to distinguish ultrasonic from audible commands. NormDetect uses an unsupervised model combined with a memory network that is lightweight and can be applied to a variety of devices. Machine learning approaches, including CNN and SVM, analyze extracted audio features, such as power and frequency patterns, to classify signals as either benign or malicious. Another technique includes a guard signal that generates a specific ultrasonic signal to interfere with and cancel out potential attack signals.

Chapter 4

Conclusion

This chapter concludes the seminar report by performing a comparison between the methods discussed in the previous chapter based on the advantages and disadvantages of the methods used.

4.1 Comparison of Existing Technologies

Paper Title	**Advantages**	**Disadvantages**
DolphinAttack: Inaudible Voice Commands	• Implements software and hardware defense • Hardware independent	• Limited by distance of 5ft • Attack sample dependent
EarArray	• Lightweight • Does not require any hardware modification	• Hardware Dependent • Attack sample dependent

Table 4.1: Comparison of methods

Paper Title	**Advantages**	**Disadvantages**
Robust Detection of Machine-induced Audio Attacks	• prevents various types of audio attacks • adopts an unsupervised domain adaptation training	• Hardware Dependent • Attack sample dependent
Canceling Inaudible Voice Commands Against Voice Control Systems	• portable device with low power consumption • entire detection and cancellation process takes only 40ms • Attack Sample independent	• Hardware Dependent
Learning Normality is Enough	• Universal & Lightweight • Hardware independent • Attack sample independent	• Ineffective for attack range of only 30 cm

Table 4.2: Comparison of methods cont.

4.2 Conclusion

Inaudible voice attacks, such as DolphinAttack, represent a serious and evolving security threat to voice-controlled systems. These attacks exploit the ultrasonic frequency range, which is undetectable to humans but can be picked up by devices like smart speakers and smartphones. As these devices become more integrated into our lives, controlling home appliances, handling sensitive personal information, and even authorizing financial transactions, the risk of malicious commands delivered through inaudible channels grows significantly. Studying inaudible voice attacks is essential to understanding and addressing the vulnerabilities inherent in voice-controlled systems, ensuring they remain secure against unauthorized access and manipulation. Research into these attacks not only strengthens the defense mechanisms but also contributes to the field of voice security. Identifying and neutralizing ultrasonic threats requires advanced detection techniques, blending machine learning, signal processing, and acoustic principles. By developing robust solutions, the resilience of voice-based interfaces against various types of attacks can be enhanced, making these systems safer for personal and commercial use. Ultimately, this research is crucial for safeguarding user privacy, maintaining device integrity, and building public trust in the rapidly growing ecosystem of voice-controlled technology.

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Appendix A: Presentation

A Software-based Mitigation against Inaudible Voice Attacks

Guided by
Dr. Anita John

Presented by
Shreya Veeraraghav
S7 CSE – Gamma
52

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Title

Learning normality is enough: a software-based mitigation against inaudible voice attacks.

Authors

Xinfeng Li, Xiaoyu Ji, Chen Yan, Chaohao Li, Yichen Li, Zhenning Zhang, and Wenyuan Xu.

Published in

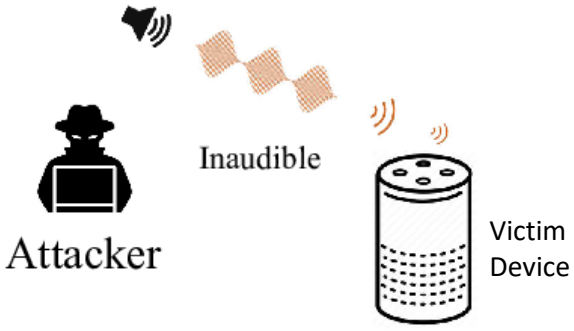
Proceedings of the 32nd USENIX Conference on Security Symposium (SEC '23). USENIX Association, USA, Article No. 138, pages 2455–2472.

2

Overview

1 Introduction
2 Motivation
3 Objective
4 Inaudible Voice Attacks
5 Attack Investigation
6 System Overview
7 Dataset
8 Overall Performance
9 Comparison with existing defenses
10 Conclusion
11 References

INTRODUCTION



- sending commands to voice-activated systems at frequencies beyond the range of human hearing
- manipulate devices that are programmed to recognize voice commands without the user being aware of it

- **Victim Devices** : Voice – activated systems (smartphones, smart speakers)
- **DolphinAttack** first demonstrated in 2017

MOTIVATION

- smartphone companies indicate software mitigations that can instantly protect miscellaneous existing devices without any hardware modification are in need
- existing devices designed with diverse features and resources - need a defense that is transferable and lightweight.
- patterns of inaudible voice attacks are diverse, benign audios share similar patterns - motivated to detect the attacks (the anomaly) by learning the benign audios (the normality).

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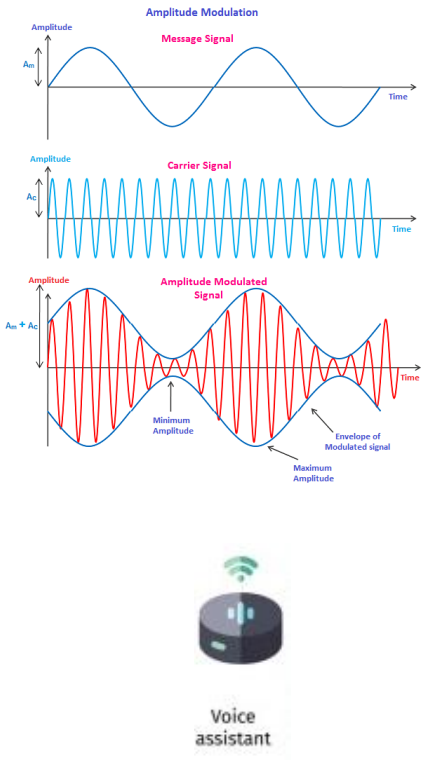
OBJECTIVE

A holistic unsupervised learning algorithm that can detect inaudible voice attack for many smart devices only based on accessible normal audios, that meets the following goals :

1. Lightweight
2. Universal
3. Unsupervised

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INAUDIBLE VOICE ATTACKS



1. Modulation : Attackers modulate a voice command to an ultrasonic frequency, by **amplitude modulation** (AM).

2. Device Reception : Voice recognition systems detect them because their microphones have wider frequency responses than the human ear.

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3. Command Execution : device interprets the modulated signal as a legitimate voice command and performs the associated action.

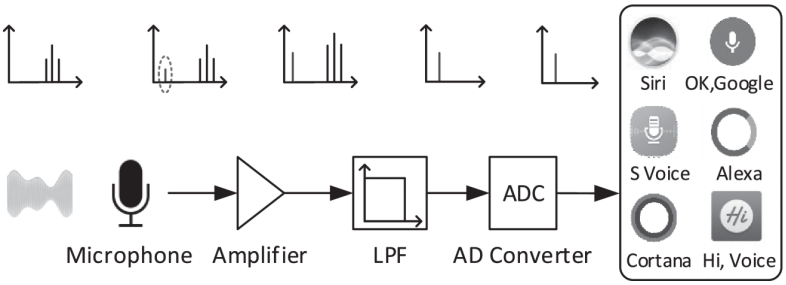


Fig. Workflow of an Inaudible Voice Attack

4. Attack Distance : attacker only needs to be within a certain distance (a few meters) from the target device to perform the attack.

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ATTACK INVESTIGATION

RQ 1: What are the differences between benign audio and inaudible voice attacks?

RQ 2: How different are the benign audio and inaudible voice attacks?

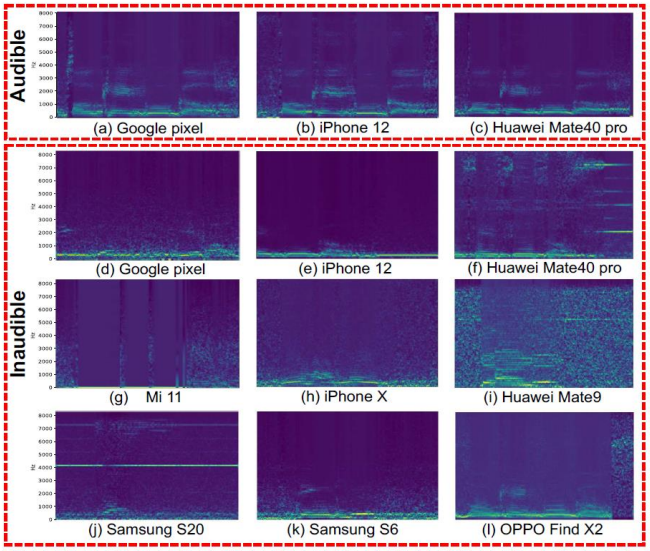
RQ 3: Why do the attack patterns vary on different target devices?

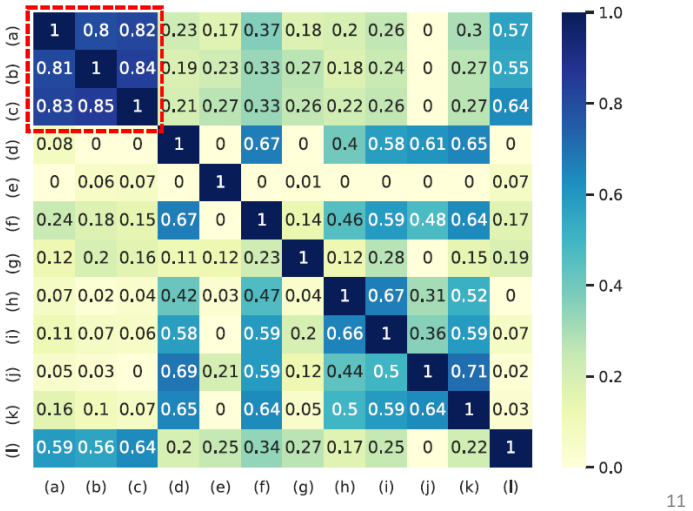
RQ1 : What are the differences between benign audio and inaudible voice attacks?

- quality of the injected attack audio varies greatly with the target device
- quality of the recorded benign voices is clear and similar across all devices
- attack power directly affects the audio quality
- speech commands have little impact on the attack effectiveness

RQ2 : How different are the benign audio and inaudible voice attacks?

- spectrums of the 3 audible cases are quite similar
- Patterns of inaudible commands are different from each other



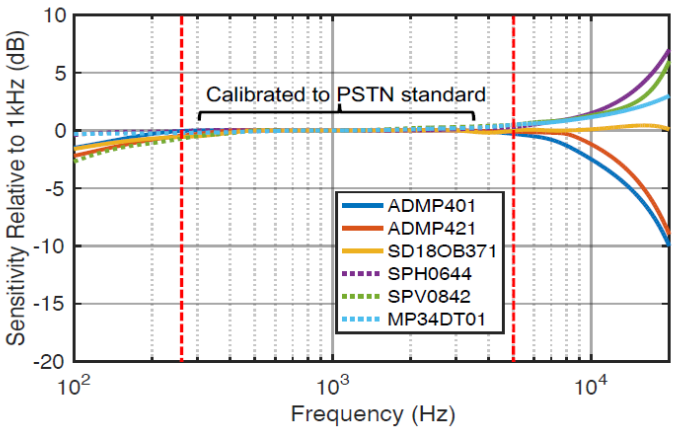


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RQ3 : Why do the attack patterns vary on different target devices?

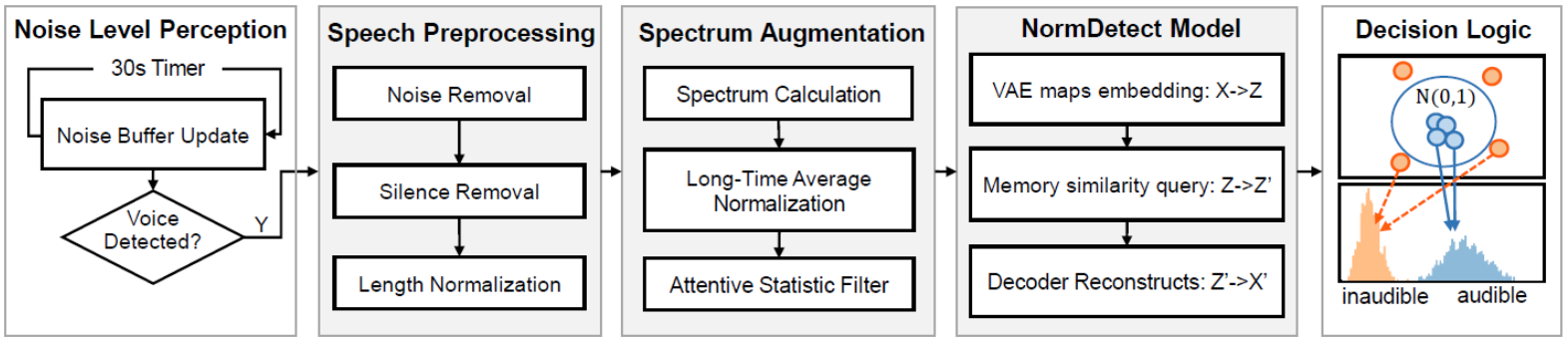
- Root cause : diverse frequency response of microphones in the upper 10 kHz band

- normal voices more similar to each other because the microphone's frequency response from 300 to 3400 Hz is close due to the manufacturers' calibration to comply with the PSTN standard.



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SYSTEM OVERVIEW

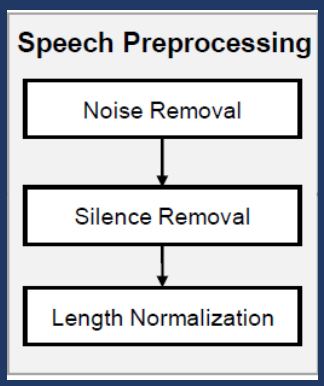


3 key stages:

1. Speech preprocessing
2. Spectrum Augmentation
3. NormDetect Model

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Speech Preprocessing



1. Noise Removal

   - since the ambient noise is somewhat similar to the abnormal microphone response caused by ultrasonic waves

2. Silence Removal

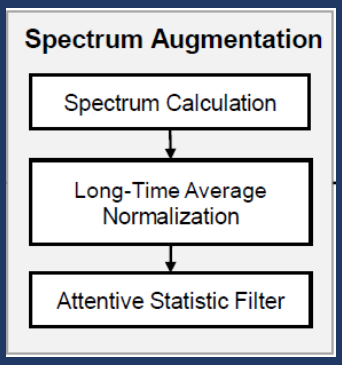
   - ensures that the model can focus on learning voiced features

3. Length Normalization

   - ensures that all voice commands, regardless of their original duration, have a consistent input length when fed into the machine learning model.

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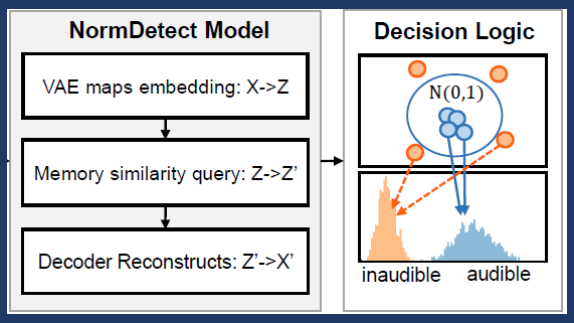
Spectrum Augmentation



1. Spectrum Calculation/Generation
   - generates the spectrum using Short-Time Fourier Transform(STFT)
2. Long-Time Average Normalization
   - generates Long-Time Average Spectrum(LTAS) to minimize variations of different speech content and speaker
3. Attentive Statistical Filter
   - enhances high discriminative information between audible and inaudible classes using the Fratio method.

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NormDetect Model



3 stages :

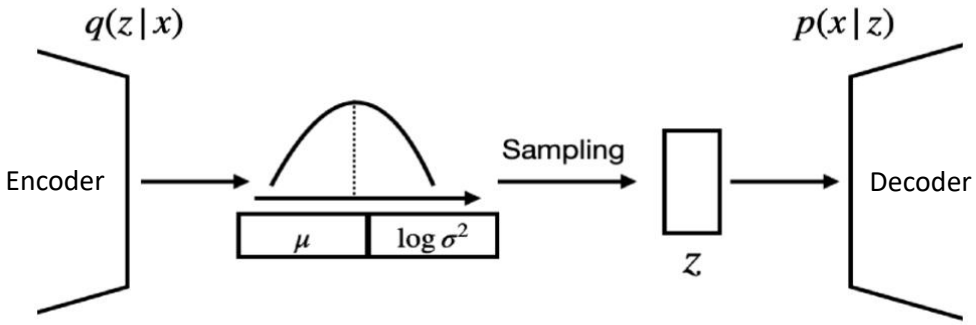
1. maps the augmented spectrums into a latent space (X -> Z)
2. queries the memory module that only stores normal embeddings to make each input similar to normal patterns (Z -> Z')
3. reconstructs the embedding to a spectrum (Z' -> X')

- Uses a Variational Autoencoder

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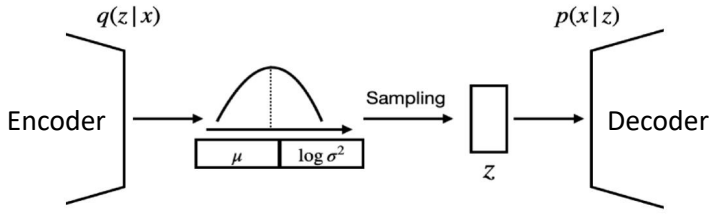
What is a Variational Autoencoder (VAE) ?

Variational autoencoders (VAEs) are generative models that learn compressed representations of their training data as *probability distributions,* which are used to generate new sample data by creating *variations* of those learned representations.



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Structure of a VAE



Encoder :

- maps input x to a Gaussian distribution over the latent space
- learns 2 variables – mean and variance, which define the Gaussian distribution for each input

Decoder :

- latent variable z is sampled from this Gaussian distribution, $z \sim \mathcal{N}(\mu(x), \sigma^2(x))$

which is used by the decoder to reconstruct the input or generate new data.

$$z = \mu + \sigma \cdot \epsilon$$

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Loss Function

$$L_{total} = L_{recon} + \beta L_{KL}$$

The training of the VAE is guided by two main loss components:

a. Reconstruction Loss (L_{recon}) :

- measures how well the decoder can reconstruct the original input **x** from the latent variable **z**.

$$L_{recon} = -\sum (x \cdot \log(\hat{x}) + (1 - x) \cdot \log(1 - \hat{x}))$$ (Cross Entropy)

b. KL(Kullback-Leibler) Divergence Loss(L_{KL}) :

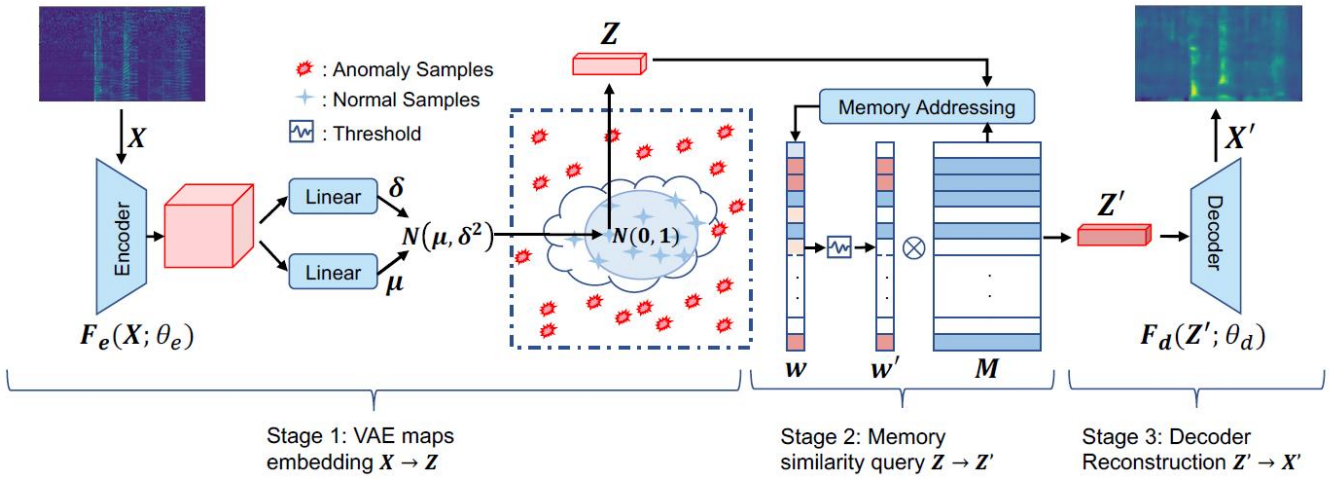
- to measure how much the learned latent distribution differs from prior distribution.

$$L_{KL} = D_{KL}\left[q_\phi(z|x) \parallel p_\theta(z)\right] = \int q_\phi(z|x) \log\left(\frac{q_\phi(z|x)}{p_\theta(z)}\right) dz$$

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NormDetect Model

Principle : the model reconstructs the normal well based on standard training data while reconstructs poor for abnormal data.



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STAGE 1: VAE maps embedding X -> Z

2 goals –

- Successful reconstruction
- Constraint embedding

- Maps the input spectrum X into a latent embedding Z following Gaussian distribution
- Reparameterization technique used to add noise so that VAE trains to reconstruct various normal patterns in a robust manner

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STAGE 2: Memory Similarity Query Z -> Z'

Objective : to constrain the generalization capability of VAE to keep the inaudible voice commands from being falsely accepted

- Memory module consists of items mi which represent patterns of normal audio commands

$$w = ZM = [Zm_1, Zm_2, \ldots, Zm_n]$$

$$Z' = w'M = \sum_{i=1}^{N} \omega_i' m_i$$

where,

w – similarity weights vector

w' – refined weights vector

Z' – adjusted embedding

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STAGE 3: Decoder reconstructs Z' -> X'

- Reconstructs spectrum X' from queried embedding Z'
- Evaluated training using Cross Entropy
- Anomaly score computed using negative log-likelihood between X and X'

$$\mathrm{NLL} = -\log(P(X|X'))$$

DATASET

Table 1: The details of two datasets in our experiment

	Speakers	Data Source	Number of Utterance	Language	Average Duration
Training Set	97	Fluent Speech Commands Dataset[9]	30,043	English	3.16s
Test Set	28	Audible & Inaudible Voice Commands Corpus	383,320	English & Chinese	1.31s

Table 6: List of Audible & Inaudible Voice Commands Dataset

Developer	English Wake-up Words	Developer	Chinese Wake-up Words	Phrases	Command
Amazon	Echo	JD	DingDongDingDong	serious everywhere	Take the picture
Amazon	Computer	MeiZu	NiHaoMeiZu	everyone hundred	Call my mom
Amazon	Amazon	Tencent	XiaoWeiXiaoWei	period anywhere	Navigate to my home
Google	OK Google	Lenovo	NiHaoLianXiang	unfortunately	Restart phone now
Google	Hey Google	Huawei	XiaoYiXiaoYi	together difficult	Turn off the light
Microsoft	Hey Cortana	Huawei	NiHaoYoYo	didn't agreement	
Samsung	Hey Bixby	Mobvoi	NiHaoWenWen	immediately connect	
Apple	Hey Siri	OPPO	XiaoBuXiaoBu		
OPPO	Hey Breeno	OPPO	NiHaoXiaoBu		

OVERALL PERFORMANCE

Evaluation Metrics

- False Rejection Rate (FRR) :
  - rate at which the system falsely rejects audible commands
- False Acceptance Rate (FAR) :
  - rate at which an inaudible voice command is wrongly accepted by the system as normal.

- A higher AUC(Area Under Curve) score indicates that the system has more discriminative power against inaudible voice attacks.

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OVERALL PERFORMANCE

Result

The model demonstrates :

- Strong overall discriminatory power
- Adaptability across devices
- Real-World Applicability

Evaluation Setting.	AUC.
All Data Mixed	99.11%
All Data Averaged	99.48%

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COMPARISON OF EXISTING DEFENSES

NAME	HARDWARE INDEPENDENT	ATTACK SAMPLE INDEPENDENT	SAMPLES	DEVICE	METHODOLOGY
Zhang et al.	✓	✗	24	1 device	SVM
LipRead	✓	✗	2368	3 devices	Elliptical Classifier
GuardSignal	✗	✓	/	1 prototype	Signal Fingerprinting, Acoustic Feature Extraction
EarArray	✗	✗	13000	1 prototype mic array	Beamforming and Microphone Array Processing
Li et al.	✗	✗	1560	3 mic arrays	Convolutional Neural Network
NormDetect	✓	✓	383320	24 devices	VAE with Memory Similarity Query

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CONCLUSION

- discovered that patterns of inaudible voice attacks are diverse and distinct from that of the audible, while audible voices share similar patterns
- designed a universal detection model NormDetect, different from the supervised learning approach of existing studies
- NormDetect performs an average AUC of 99.48% , suggesting its effectiveness in detecting inaudible voice attacks.

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References

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3. Hadi Abdullah, Kevin Warren, Vincent Bindschaedler, Nicolas Papernot, and Patrick Traynor. Sok: The faults in our asrs: An overview of attacks against automatic speech recognition and speaker identification systems. In 2021 IEEE symposium on security and privacy (SP), pages 730–747. IEEE, 2021.
4. Guoming Zhang, Chen Yan, Xiaoyu Ji, Tianchen Zhang, Taimin Zhang, and Wenyuan Xu. Dolphinattack: Inaudible voice commands. In Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, pages 103–117, 2017.

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THANK YOU

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Questions & Answers

1. Why is amplitude modulation preferred over frequency modulation for producing the ultrasonic attack signal?

   Answer: Amplitude modulation(AM) is easier to implement in both the transmitter and receiver in terms of circuitry and signal generation, whereas Frequency modulation(FM) requires complex circuits to modulate a signal on an ultrasonic carrier, which is less-efficient for high-frequency signals.

   Also, AM signals can be detected using simpler demodulation techniques, which is computationally less intensive and easier to implement in the receiver, while FM requires sophisticated techniques.

2. What is the advantage of using an unsupervised algorithm?

   Answer: Supervised learning algorithms require a large, labeled dataset that consist of attack samples in addition to normal samples, and obtaining of such attack signals is difficult, and hence these signals must be generated with costly harware in order to train the model. On the other hand, unsupervised algorithms don't require labeled data. They learn the normal behavior or patterns of the input, and can detect deviations or anomalies from the learned patterns.

3. What do mean and variance of a gaussian distribution for an input data indicate?

   Answer: The mean represents the point in the latent space defined by the gaussian distribution that the VAE considers the best representation of the input data x, capturing the core features. The variance represents how uncertain the VAE is, about the encoding of x in the latent space. A low variance means that the model is very sure in its representation of x, whereas a high variance indicates more uncertainity.

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and Team work: Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

10. Communication: Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

After the completion of the course the student will be able to:

Course Outcome 1: Identify academic documents from the literature which are related to her/his areas of interest (Cognitive knowledge level: Apply).

Course Outcome 2: Read and apprehend an academic document from the literature which is related to his/her areas of interest (Cognitive knowledge level: Analyze).

Course Outcome 3: Prepare a presentation about an academic document (Cognitive knowledge level: Create).

Course Outcome 4: Give a presentation about an academic document (Cognitive knowledge level: Apply).

Course Outcome 5: Prepare a technical report (Cognitive knowledge level: Create).

Appendix C: CO-PO-PSO Mapping

CO - PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1	2	2	1	1		2	1					3
2	3	3	2	3		2	1					3
3	3	2			3			1		2		3
4	3				2			1		3		3
5	3	3	3	3	2	2		2		3		3

CO - PSO Mapping

CO	PSO 1	PSO 2	PSO 3
1	3	1	2
2	3	2	2
3	2	2	
4	3		3
5	3		

Justification

Mapping	Justification
CO1 - PO1	Apply suitable computer science theory and mathematical foundations to identify relevant research areas
CO1 - PO2	Analyse the different problems and select relevant literature
CO1 - PO3	Create explainable content about the concepts to be conveyed to the audience
CO1 - PO4	Apply mathematical foundations and computer science theory to solve technical problems in the design of computer-based systems
CO1 - PO6	Apply multidisciplinary knowledge in projects, utilizing suitable computer science theory and mathematical foundations in team-oriented tasks
CO1 - PO7	Identify solutions from literature that can lead to a sustainable solution
CO1 - PO12	Analyze the recent and relevant works done in their area of research to understand the methodologies
CO2 - PO1	Analyze academic documents using mathematical foundations and algorithm principles relevant to the modeling and design of computer-based systems.
CO2 - PO2	Analyze and comprehend academic documents from the literature related to computer-based system modeling and design
CO2 - PO3	Evaluate and analyse computing requirements for technical programs in relevant literature chosen and understand them
CO2 - PO4	Identify the new technologies and modern engineering tools used and understand their working
CO2 - PO6	Apply comprehension from the literature to assess health, societal and legal issues and their relation to the engineering problems.
CO2 - PO7	Read and apply the context from the literature for sustainable development
CO2 - PO12	Ability to read and summarize the developments in engineering for lifelong learning process
CO3 - PO1	Create presentations applying mathematical foundations and algorithmic principles to communicate the design of computer-based systems
CO3 - PO2	Present academic documents effectively, applying computing and mathematical principles for problem analysis
CO3 - PO5	Present academic documents effectively, applying computing and mathematical principles in problem-solving
CO3 - PO8	Present multidisciplinary knowledge effectively,applying computing and mathematical principles in seminar presentations
CO3 - PO10	Present academic documents effectively, demonstrating communication skills
CO3 - PO12	Present the concepts and topics under study effectively

Mapping	Justification
CO4 - PO1	Apply mathematical foundations and algorithmic principles to give presentations about computer-based system modeling
CO4 - PO5	Give presentations about academic documents, applying modern tools and techniques
CO4 - PO8	Give presentations about academic documents keep up engineering norms and ethics
CO4 - PO10	Presenting the seminar strengthens the knowledge of the topic and promotes communication skills
CO4 - PO12	Presenting the seminar strengthens the knowledge of the topic and promotes life-long learning for continuous professional development
CO5 - PO1	Prepare a technical report and apply mathematical foundations in designing computer systems
CO5 - PO2	Enhances problem-solving for formulating technical solutions
CO5 - PO3	Develops skills in defining computing requirements and system design
CO5 - PO4	Prepare a technical report for experiments on complex problems
CO5 - PO5	Promotes experimentation and analysis for evidence-based reports
CO5 - PO6	Prepare technical reports that can be useful for the society as well
CO5 - PO8	Prepare a technical report and apply mathematical foundations in designing computer systems making sure that publication ethics are met
CO5 - PO10	Sets professional, ethical, and legal principles in technical reports
CO5 - PO12	Record the academic reports and concepts for life-long learning
CO1-PSO1	Students will be able to apply their skills to identify, analyze, and formulate solutions for complex, multidisciplinary engineering problems through the strategic use of academic literature.
CO1-PSO2	Students will be able to navigate academic literature effectively while applying programming concepts, thereby preparing them for more advanced learning and practical applications in the field of computer science.
CO1-PSO3	Students will demonstrate their ability to interpret and utilize fundamental computer science concepts to conduct competitive research through academic literature.
CO2-PSO1	Students will be able to evaluate and design solutions for complex engineering problems in multidisciplinary areas by critically analyzing and interpreting academic documents from the literature.
CO2-PSO2	Students will be able to locate and utilize academic literature effectively while applying programming concepts, thereby equipping them for further learning and practical applications in the field of computer science.
CO2-PSO3	Students will have the ability to understand and apply the fundamentals of computer science in competitive research by interpreting and utilizing academic documents from the literature.

Mapping	Justification
CO3- PSO1	Students will be able to identify, analyze, and develop solutions for complex engineering problems across multidisciplinary areas while creating a presentation on an academic document
CO3-PSO2	Students will demonstrate the ability to apply the fundamentals of computer science in competitive research by designing a presentation based on an academic document
CO4- PSO1	Students will be able to to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas while giving a presentation about an academic document
CO4- PSO3	Students will have the ability to apply the fundamentals of computer science in competitive research while giving a presentation about an academic document
CO5- PSO1	Students will be able to to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas while preparing a technical report